

# Coding & Solution

|               |                                                         |
|---------------|---------------------------------------------------------|
| Date          | 30 September 2026                                       |
| Team ID       | SWTID-2026-7446                                         |
| Project Name  | ComicCraft - AI Comic Story Creator using Gemini Models |
| Maximum Marks | 5 Marks                                                 |

## Coding & Solution

This section evaluates the quality and completeness of the implemented ComicCraft application. The solution uses FastAPI, Google Gemini models, Stable Diffusion, Jinja2 templates, and FPDF to provide an end-to-end workflow for AI-powered comic creation.

### Instructions

1. Ensure the code is well-structured, modular, commented where necessary, and follows suitable Python/FastAPI practices.
2. Implement the core ComicCraft workflow: prompt collection, panel outline generation, story narration and dialogue, image generation, layout construction, and PDF export.
3. Include source files, configuration/dependency files, environment-variable setup, and local run instructions.
4. Attach or link the code repository in the repository field below when available.

### Solution Summary

| Field                         | Details                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Repository Link / URL         | [Paste GitHub / GitLab or hosted repository link here]                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Programming Language(s)       | Python, HTML, CSS                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Framework(s) Used             | FastAPI, Jinja2                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| AI / ML Technologies          | Google Gemini Flash, Google Gemini Pro, Hugging Face Diffusers, Stable Diffusion, PyTorch                                                                                                                                                                                                                                                                                                                                                                                        |
| Key Features Implemented      | <ul style="list-style-type: none"><li>• User input for story prompt, character name, setting, tone, and art style</li><li>• 5-panel structured comic outline generation</li><li>• Detailed narration and character dialogue generation</li><li>• Comic-style image generation for each panel</li><li>• Panel layout construction</li><li>• Multi-page PDF export</li><li>• Web preview and export-success flow</li><li>• JSON API support through /generate-comic/json</li></ul> |
| Pending / Incomplete Features | The documented version focuses on single-comic generation. Future enhancements may include user profiles, comic libraries, and multi-page story arcs.                                                                                                                                                                                                                                                                                                                            |
| Setup / Run Instructions      | <ol style="list-style-type: none"><li>1. Create and activate a Python virtual environment.</li><li>2. Install dependencies using requirements.txt or the documented pip packages.</li><li>3. Configure GEMINI_API_KEY and HF_API_KEY environment variables.</li><li>4. Run: <code>uvicorn app.main:app --reload</code></li><li>5. Open <code>http://127.0.0.1:8000</code></li><li>6. API documentation: <code>http://127.0.0.1:8000/docs</code></li></ol>                        |

### Code Quality Checklist

| S.No | Criteria                                               | Status (Yes / No) |
|------|--------------------------------------------------------|-------------------|
| 1    | Code is modular and organized into functions / classes | Yes               |
| 2    | Meaningful variable and function names are used        | Yes               |
| 3    | Code includes comments / documentation where necessary | Yes               |
| 4    | Error handling is implemented for critical operations  | Yes               |
| 5    | The application runs without critical errors           | Yes               |
| 6    | Code is committed to a version control repository      | To be confirmed   |

### Additional Notes / Comments

ComicCraft provides an end-to-end AI comic generation pipeline. Gemini Flash is used for structured panel outlines, Gemini Pro for detailed narration and dialogue, and Stable Diffusion for illustrations. The generated panels are assembled into a layout and exported as a downloadable PDF. The architecture separates frontend templates, FastAPI routing, AI integrations, layout building, and exporting.

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